CSE7302- University Project-Review 2

PROJECT TITLE—"AI-Powered Multi-Scan Medical Imaging and Report Interpretation System."

Batch Number: CSE-115

ROLL NUMBER	NAME
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20221CSE0021	FAIZAN AHMED
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Under the Supervision of,

Mr. Praveen Giridhar Pawaskar
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Name of the Program: Computer Science and Engineering
Name of the HoD: Blessed Prince
Name of the Program Project Coordinator: Muthuraj V
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Abstract

- Medical imaging plays a critical role in early disease detection and clinical decision-making.
- However, fragmented imaging workflows, delayed report interpretation, and lack of multi-scan correlation reduce diagnostic efficiency.
- ScanWise proposes an AI-powered software platform that integrates multiple medical scans (X-ray, CT, MRI, Ultrasound) and radiology reports into a unified intelligent analysis system.
- The system leverages deep learning, computer vision, and agentic AI to assist clinicians with preliminary interpretation, anomaly highlighting, and patient-friendly summaries.
- The solution aims to improve diagnostic speed, reduce clinician workload, and enhance patient understanding while maintaining clinical safety.

Literature Survey Summary%@

Key Reviewed Papers:

1. **Comprehensive Review:** Analyzed **10 relevant research papers** from IEEE Xplore
2. **Key Finding:** Traditional dispatch systems rely heavily on manual decision-making, leading to delays and suboptimal resource allocation.
3. **Opportunity:** Advancements in **Machine Learning**, server-side rendering, and real-time data processing offer significant opportunities for improvement.
4. **Our Focus:** Applying structured **AI flows** for incident classification and using **predictive analytics** for demand forecasting.

Literature Survey - Comprehensive Review & Critical Analysis

Objectives

- To develop an AI-based system for multi-scan medical image analysis.
- To integrate radiology reports with image-based findings.
- To assist clinicians with early anomaly detection and triage support.
- To generate simplified patient-friendly explanations.
- To ensure scalability, security, and cloud deployment feasibility.

Existing Methods and Drawbacks

- **Existing Methods**

Manual radiologist interpretation
Single-modality AI diagnostic tools
Standalone PACS and RIS systems
Rule-based clinical decision support

- **Drawbacks**

Time-consuming and workload-intensive
High inter-observer variability
No cross-scan correlation
Poor patient engagement
Limited automation and scalability

Proposed Method

- AI-powered multi-modal medical imaging platform
- Supports X-ray, CT, MRI, and Ultrasound
- Combines computer vision + NLP + agentic AI

Provides:

- Preliminary AI insights for clinicians
- Confidence-based anomaly highlighting
- Simplified explanations for patients

Feasibility Study

Technical Feasibility:

- Proven deep learning architectures (CNNs, Vision Transformers)
- Cloud-native deployment using scalable infrastructure
- Secure healthcare data handling

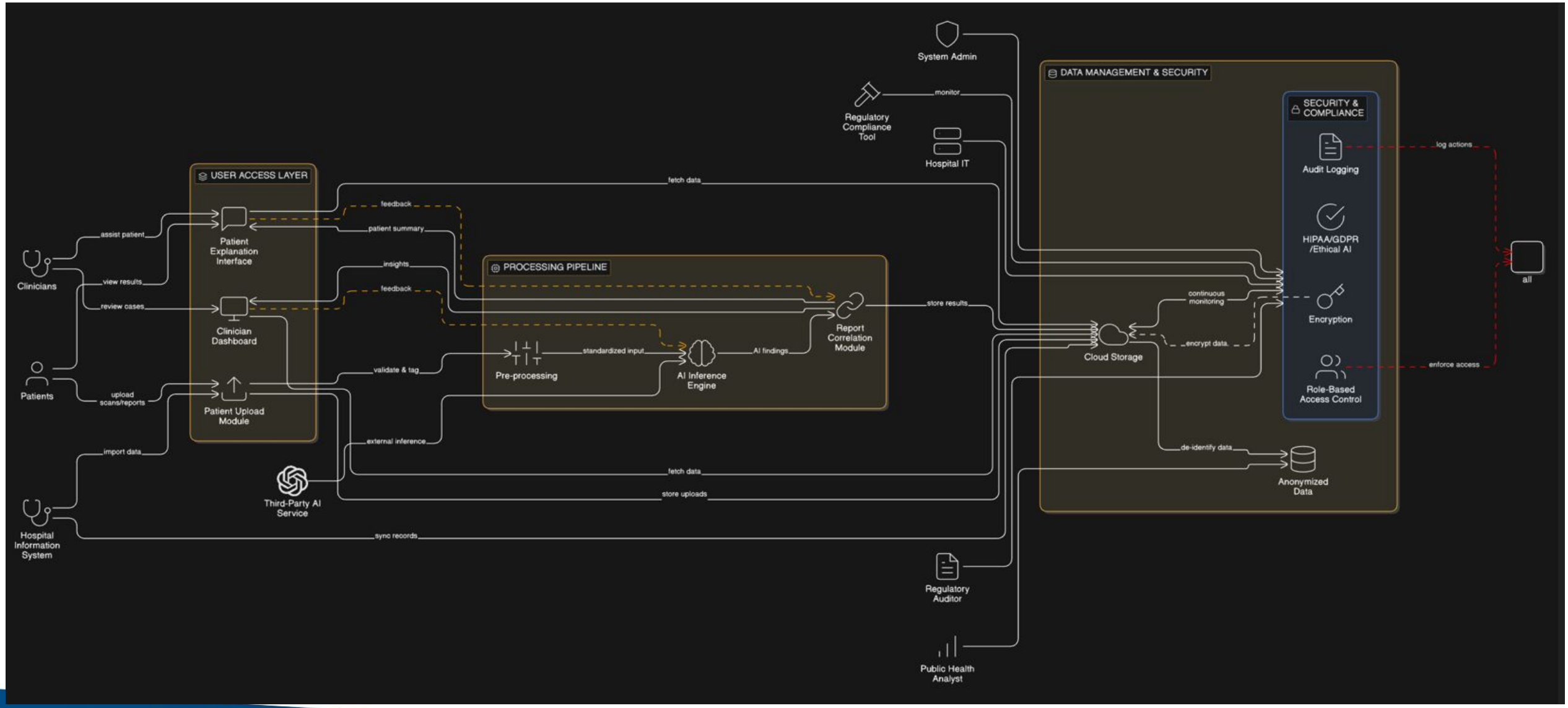
Economic Feasibility:

- Software-only solution
- No additional hardware requirements
- Open-source frameworks reduce cost

Operational Feasibility

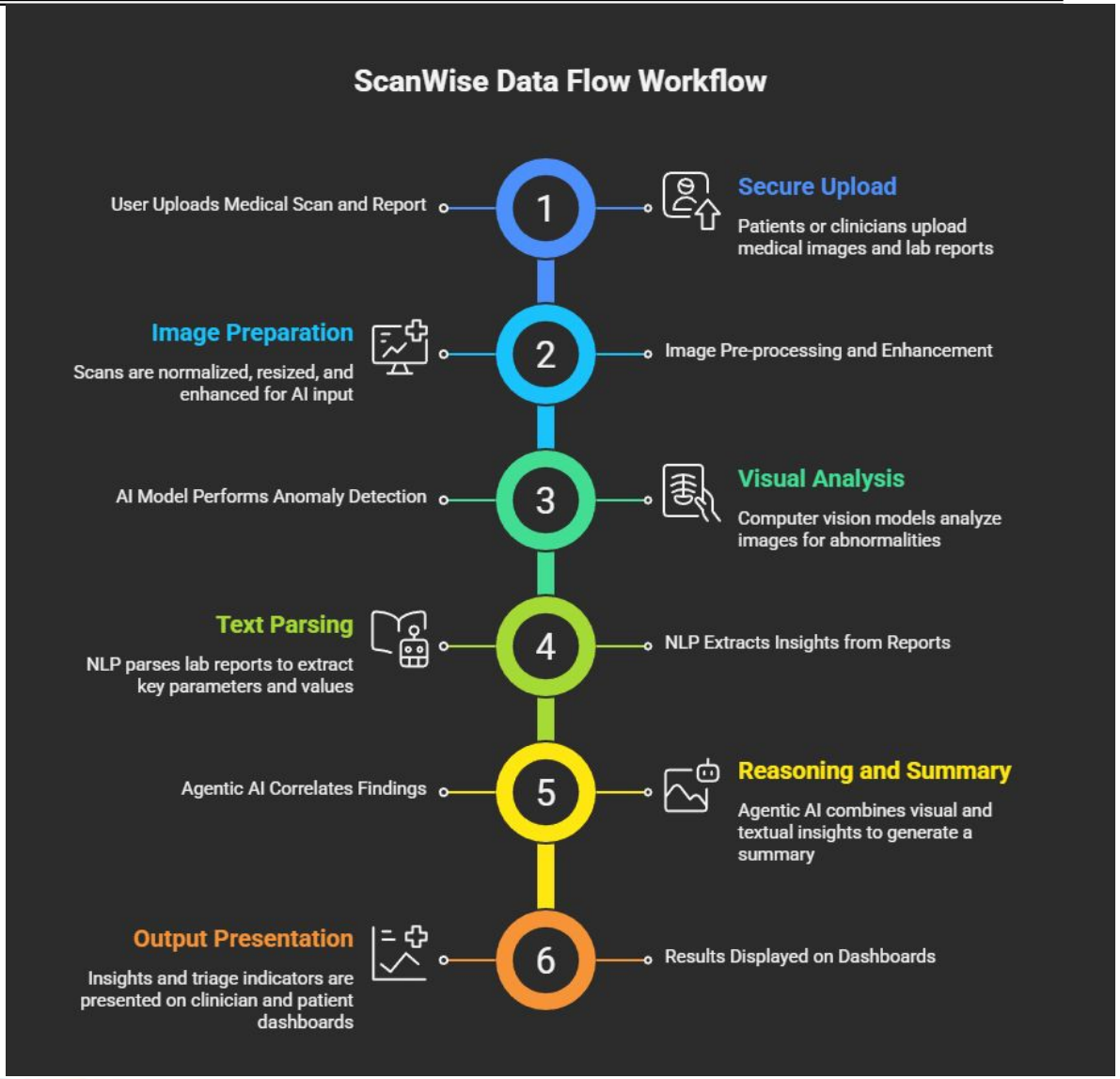
- Integrates with existing hospital workflows
- Minimal learning curve for users

System Architecture Diagram



Modules

- Scan Ingestion Module
- Image Pre-Processing Module
- AI Diagnosis Module
- Report Interpretation Module
- Agentic Decision Support Module
- Clinician Dashboard
- Patient Communication Module



Technology Stack

Component	Technology
Frontend	React + TypeScript
Backend	Node.js + TypeScript
AI Engine	Gemini (Vision + LLM)
Database	Firestore
File Storage	Firebase Auth
Authentication	Firebase Auth
Maps	Google Maps API

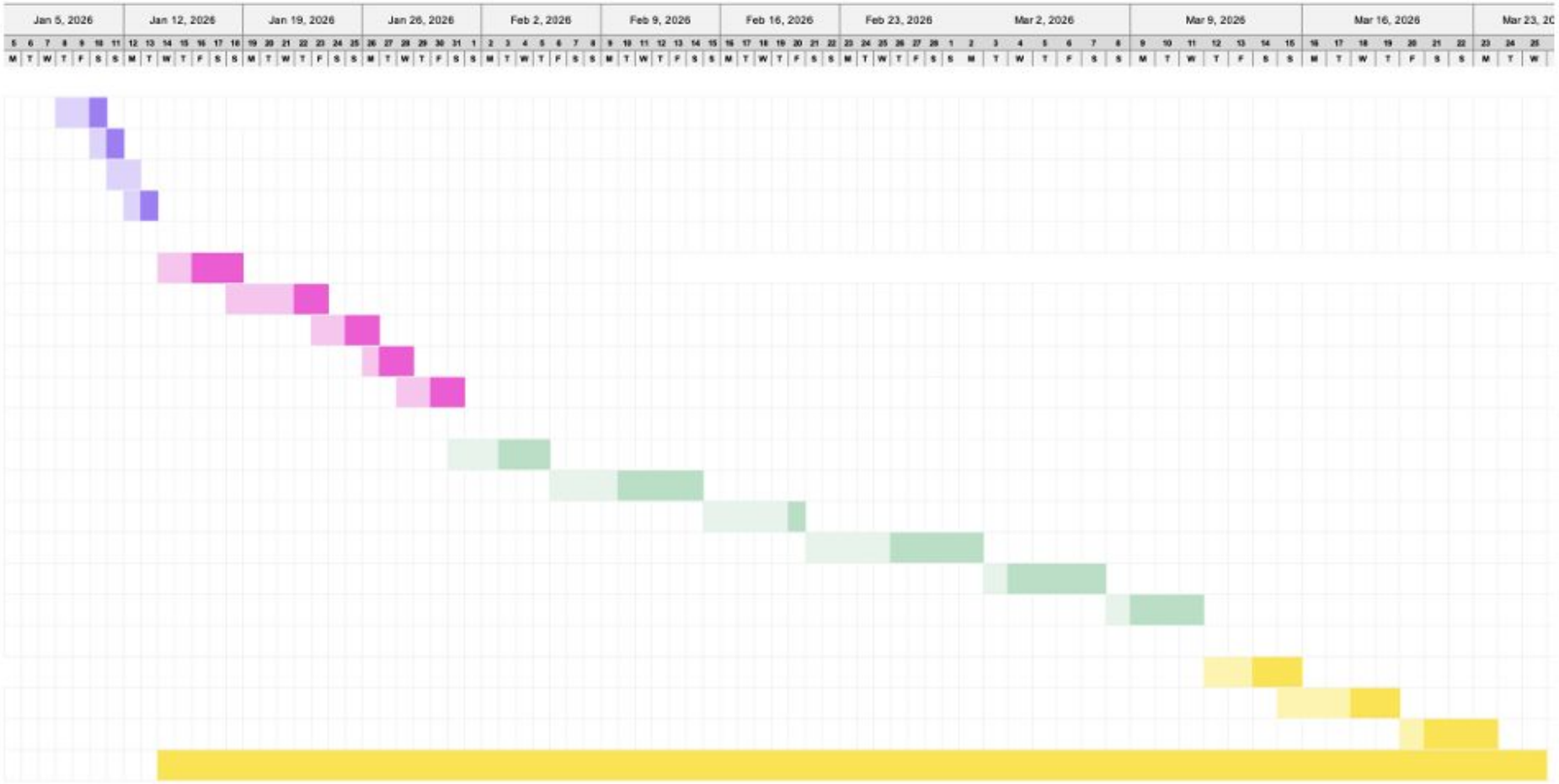
Timeline of the Project (Gantt Chart)

ScanWise

Presidency University

Project start: 8/1/26

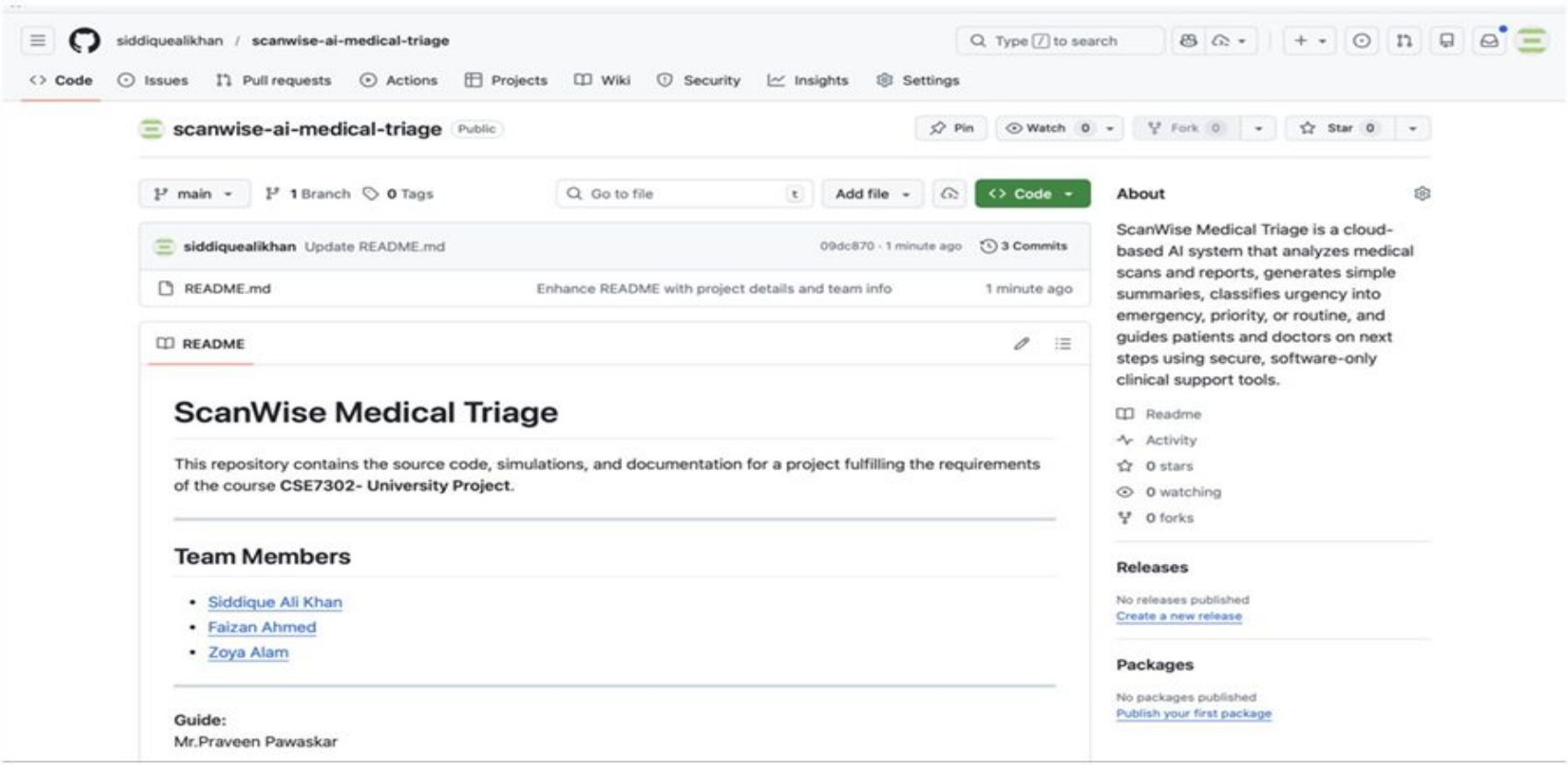
TASK	ASSIGNED TO	START	END
Initiation			
Define problem statement	Siddique Ali Khan	8/1/26	10/1/26
Literature survey	Faizan Ahmed	10/1/26	11/1/26
Requirement analysis	Zoya Alam	11/1/26	12/1/26
Tool & tech selection	Siddique Ali Khan	12/1/26	13/1/26
Planning and design			
System architecture design	Siddique Ali Khan	14/1/26	18/1/26
UI/UX wireframe design	Faizan Ahmed	18/1/26	23/1/26
Database schema design	Zoya Alam	23/1/26	26/1/26
API & backend flow design	Siddique Ali Khan	26/1/26	28/1/26
AI pipeline design	Faizan Ahmed	28/1/26	31/1/26
Execution			
Frontend development (React)	Faizan Ahmed	31/1/26	5/2/26
Backend development (Node/Express)	Siddique Ali Khan	6/2/26	14/2/26
Database & Firebase setup	Zoya Alam	15/2/26	20/2/26
AI model integration	Faizan Ahmed	21/2/26	2/3/26
Triage logic implementation	Zoya Alam	3/3/26	7/3/26
Authentication, security & system integration	Siddique Ali Khan	8/3/26	11/3/26
Evaluation			
Module testing	Zoya Alam	12/3/26	15/3/26
UI testing & fixes	Siddique Ali Khan	15/3/26	19/3/26
System testing	Faizan Ahmed	20/3/26	23/3/26
Report writing	Everyone	14/1/26	25/3/26



Github Link

Repository Name: scanwise-ai-medical-triage

Public Link: https://github.com/siddiquealikhan/scanwise-ai-medical-triage



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Thank
You!